TORQUE ON A ROD OR A CYLINDER

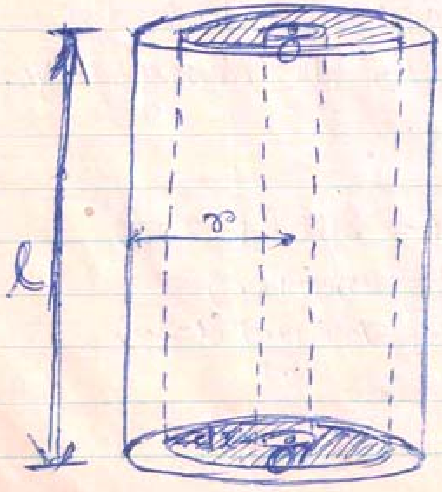
Let us consider a wire or a rod, clamped at one end and twisted at its free end, by a very small angle say ϕ. Due to this twist the length and diameter of the rod does not change only the shape changes by a very small amount.

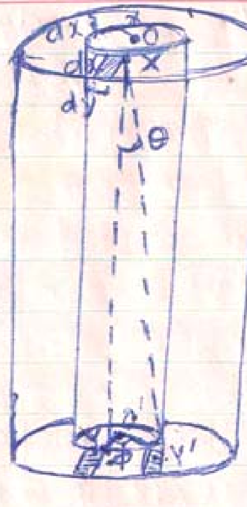
The rod can be assumed to be made up of large number of thin circular disc, kept one over the other. When force is applied parallel to the layers, the layers suffer relative angular displacement. The displacement is maximum for the layer at free end and decreases as we go towards the clamped end and is zero for the layer at clamped end. Hence this is an example of pure shear.

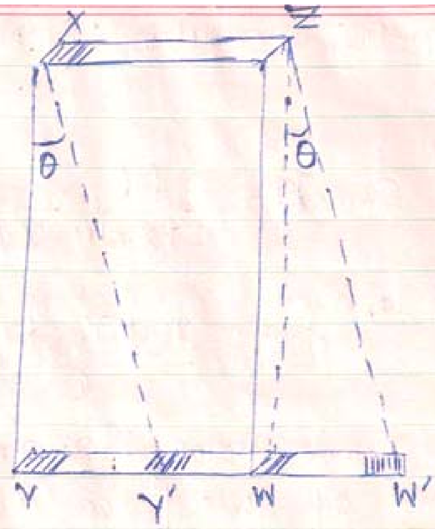
Given : l = The length of the rod.
r = The radius of the rod
η = The rigidity modulus of the rod.
ϕ = The small angle of twist at the free end.

In the fig; the rod has been shown magnified.

We can imagine the rod to be made up of large no. of hollow co-axial cylinders. Let us consider one such cylinder of radius x $(0 \leq x \leq r)$ and radial thickness 'dx' this is also shown magnified.







let us consider an elementary surface of area $dx \cdot dy$ on the upper rim at X, of the imaginary cylinder. Y is the corresponding position of the element in the lower rim, before twist. After the free end is twisted Y goes to Y' by an angle ϕ but X remains fixed in position. Hence if the cylinder were to cut along the line XY and flattened. It would have given a shape of plate rectangle XYWZ. But after the twist we get a parallelogram $X\hat{Y'}W'Z$

Hence $\hat{YXY'} = \theta$ = angle of shear or shearing strain

from $\triangledown$ YOY', $\phi = \dfrac{\text{arc } YY'}{x}$ or $\boxed{\text{arc } YY' = x\phi}$ → (1)

from $\triangledown$ YXY', $\theta = \dfrac{\text{arc } YY'}{l}$ or $\boxed{\text{arc } YY' = l\theta}$ → (2)

Equating (1) and (2) :-

$l\theta = x\phi$ or $\boxed{\text{Shearing strain } \theta = \dfrac{x\phi}{l}}$ → (3)

Thus max^m strain is produced at the outermost part & least on the innermost hence shearing strain is not uniform throughout.

let f = force on the element $dx\,dy$ = ??

Shearing stress = $\dfrac{f}{dx \cdot dy}$

③

$\therefore$ Rigidity modulus : $\eta = \dfrac{\frac{f}{dx\,dy}}{\theta} = \dfrac{\frac{f}{dx.dy}}{\frac{x\phi}{l}}$

or $\eta = \dfrac{f\,l}{dx.dy.x\phi}$

or $f = \dfrac{\eta\phi x\,dx.dy}{l}$

This force being tangential to the surface
Moment of this force about the axis of rotation

$$= f.x = \frac{\eta\phi x^2\,dx.dy}{l}$$

(On the element of length dy and thickness dx).

The total moment on the imaginary cylinder

$$= \frac{\eta\phi x^2\,dx}{l}\int dy \qquad \left[\int dy = 2\pi x\right]$$

$$= \frac{2\pi\eta\phi x^3\,dx}{l} \longrightarrow ④$$

The total moment on the rod can be obtained by integrating equation ④ between the limits $x=0$ to $x=r$

Torque on the rod (couple) $\tau = \displaystyle\int_0^r \frac{2\pi\eta\phi x^3\,dx}{l}$

$$= \frac{2\pi\eta\phi}{l}\int_0^r x^3\,dx$$

$$= \frac{2\pi\eta\phi}{l}\left[\frac{x^4}{4}\right]_0^r$$

$$= \frac{\cancel{2}\pi\eta\phi}{l}\frac{r^4}{\cancel{4}\,2} = \frac{\pi\eta r^4}{2l}\phi$$

Or $\tau = \frac{\pi \eta r^4}{2l} \cdot \phi$ ------ (5)

Equation (5) gives the torque required to twist the rod, by an angle ϕ and is known as <u>torsional couple</u>.

Since for a given rod, l, r and η are all constants, Put $\frac{\pi \eta r^4}{2l} = C = \text{Const.}$ ⟶ (6).

$\therefore \tau = C\phi$

If $\phi = 1$ radian.

$$\boxed{\tau = C}$$

Hence the Constant C can be defined as the torsional couple per unit twist.